

^{35}P β^- decay (47.3 s) 1986Wa22

Parent: ^{35}P : $E=0$; $J^\pi=1/2^+$; $T_{1/2}=47.3$ s 8; $Q(\beta^-)=3988.4$ 19; $\% \beta^-$ decay=100

^{35}P - J^π , $T_{1/2}$: From Adopted Levels of ^{35}P .

^{35}P - $Q(\beta^-)$: From 2021Wa16.

1986Wa22: ^{35}P was produced via the $^{36}\text{S}(t,\alpha)^{35}\text{P}$ reaction. 81% enriched ^{36}S targets were bombarded with 3.4-MeV tritons from the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. γ rays were detected using an intrinsic coaxial Ge detector. Measured E_γ , I_γ . Deduced levels, decay branching ratios, and $\log ft$. Compared with shell model calculations.

1972Ap01: ^{35}P was produced via the $^{37}\text{Cl}(t,\alpha p)^{35}\text{P}$ reaction. LiCl and NaCl targets were bombarded with 16-MeV tritons from the Los Alamos tandem Van de Graaff facility. γ rays were detected using an 85 cm³ Ge(Li) detector. β particles were detected using an end-window gas-flow proportional counter. Measured E_γ , I_γ , and I_β . Deduced ^{35}P $T_{1/2}$, decay branching ratios, and $\log ft$. Compared with shell model calculations.

1972Go31: ^{35}P was produced via the $^{18}\text{O}(^{19}\text{F},2p)^{35}\text{P}$ and $^{36}\text{S}(t,\alpha)^{35}\text{P}$ reactions at the second tandem of the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. ^{35}P β -delayed γ rays were detected using a 60 cm³ Ge(Li) detector. β particles were detected in coincidence with the 1572 γ using a NE102 scintillator. Measured E_γ , I_γ , and E_β . Deduced ^{35}P $T_{1/2}$, β end-point energy, mass, decay branching ratios, and $\log ft$. Compared with shell model calculations.

1971Gr53: ^{35}P was produced via the $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ reaction. 6-g $\text{NH}_4\text{Cl}(\text{nat})$ samples were irradiated with γ rays from a bremsstrahlung spectrum at the Mainz electron linear accelerator. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured E_γ . Deduced $T_{1/2}$ of ^{35}P and levels in ^{35}S .

There is a 1-MeV gap between the highest observed level at $E=2938.4$ and $Q(\beta^-)$ value=3988.4 19 (2021Wa16). There may be missing weak transitions from unobserved levels in the gap. Upper limits for the unobserved β feedings:

1991: <0.08 (1986Wa22), <2.5 (1972Go31), <0.09 (1972Ap01);

2348: <0.08 (1986Wa22), <5 (1972Go31), <0.24 (1972Ap01);

2717: <0.03 (1986Wa22), <7 (1972Go31), <0.38 (1972Ap01);

3421: <0.02 (1986Wa22);

3558: <0.11 (1986Wa22);

3597: <0.02 (1986Wa22);

3675: <0.02 (1986Wa22);

3802: <0.03 (1986Wa22).

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	$3/2^+$	87.61 d 24
1572.290 29	$1/2^+$	2.29 ps 14
2938.4 4	$(3/2)^+$	

[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

 β^- radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^-^\ddagger$	$\log ft$	Comments
(1050.0 22)	2938.4	0.47	5.0	$I\beta^-$: others: <8 (1972Go31), <0.45 (1972Ap01).
(2416.1 22)	1572.290	99.2	4.1	$I\beta^-$: other: 99 +1-7 from measured γ/β ratio (1972Ap01), 100 (1972Go31).
(3988.4 [#] 24)	0	<0.73	>7.2	$I\beta^-$: converted by evaluators from the shell-model calculated 0.73 in 1986Wa22. Others: <9 in 1972Ap01; not given in 1972Go31.

[†] From I_γ intensity balance for excited levels.

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

^{35}P β^- decay (47.3 s) 1986Wa22 (continued) $\gamma(^{35}\text{S})$

I_γ normalization: from $\Sigma\%I(\gamma \text{ to g.s.})=99.635$ deduced from $100-\%I\beta^-(\text{g.s.})$, where $\%I\beta^-(\text{g.s.})=0.365$ is converted by evaluators from the shell-model calculated $\%I\beta^-(\text{g.s.})=0.73$ in 1986Wa22. Others: $\%I\beta^-(\text{g.s.})<9$ from 1972Ap01; $\%I\beta^-(\text{g.s.})$ is not given in 1972Go31. Given that the ground state β^- -feeding has not been experimentally observed, the evaluators opt to convert the shell-model calculated value to an upper limit when deducing the normalization factor.

$E_\gamma^\dagger$	$I_\gamma^{\dagger\#}$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	$\alpha^\@$	Comments
1572.252 29	100	1572.290	1/2 ⁺	0	3/2 ⁺	(M1)	0.0001007 14	$\%I_\gamma=99.2$ E_γ : weighted average of 1571.8 2 (1971Gr53), 1572.2 4 (1972Ap01), 1572.36 24 (1972Go31), 1572.47 25 (1972Go31), and 1572.256 24 (1986Wa22).
2938.29 40	0.476 30	2938.4	(3/2) ⁺	0	3/2 ⁺			$\%I_\gamma=0.47$

† From 1986Wa22, unless otherwise noted.

‡ From the Adopted Gammas.

$^\#$ For absolute intensity per 100 decays, multiply by 0.992.

$^\@$ Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multiplicities, and mixing ratios, unless otherwise specified.

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Decay Scheme

Intensities: I_γ per 100 parent decays

Legend

